DIALGA: A Factorized, Camera-Aware Video Embedding with a Spatial Dynamics Code

Anonymous authors
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Abstract

Powerful pretrained video VAEs compress video into rich but entangled latents: object identity, object motion, and camera motion are fused into one code that cannot be read, edited, or predicted factor by factor. Recovering such structure normally means training a new VAE from scratch. We present DIALGA, a lightweight adapter that re-encodes the latent of any frozen video VAE into named, independently accessible slots — a spatial static code (identity), a per-frame spatial dynamics code (object motion), and a camera code — with a decoder structurally prevented from mixing them, and no VAE retraining. Carrying this factorization to real moving-camera robot video exposed a general failure: a global per-frame dynamics vector reconstructs real textured video 17 dB below the frozen-VAE ceiling, because the decoder can only broadcast it uniformly across space. Giving the dynamics code a spatial axis raises held-out reconstruction by +4.4 dB, which a matched-rate ablation attributes to spatial structure (+2.7 dB) rather than added rate (+0.6 dB). We show the code is cleanly disentangled (a diagonal-dominant cross-probe), controllable (identity/motion swap), and camera-aware (a known-pose path for viewpoint stability that entangled and camera-blind baselines lack), and that its dynamics code transfers to robot action read-out. We are explicit about what DIALGA is not: at extreme compression it does not beat foundation encoders on semantic accuracy or classical codecs on rate–distortion — we report both. Its contribution is a cheap, camera-aware disentanglement of frozen video-VAE latents that entangled predictive VAEs and camera-blind decomposition VAEs do not provide.

1 Introduction

A video representation should keep its kinds of information in different places: identity — what the objects are, constant over time — object motion — how they move — and, for a moving camera, camera motion — how the viewpoint moves. Modern video VAEs do the opposite. Trained to reconstruct, they pack texture, identity, object motion, and camera motion into a single unstructured latent grid that offers no axis along which a downstream user can read out identity, freeze dynamics, or subtract the camera (Tong et al., 2022; Bardes et al., 2024). That structure can be recovered, but expensively: sequential VAEs that split content from motion do so by training a VAE from scratch (Zhu et al., 2020; Bai et al., 2021), and recent large predictive video VAEs enrich the latent's motion-awareness but leave it entangled, and cost a full pretraining run (Zhao et al., 2026). We ask a cheaper question: given an already-trained, frozen video VAE, can we recover a named, factorized, camera-aware decomposition of its latent without retraining it?

DIALGA is a lightweight adapter that re-encodes each short chunk of a frozen video-VAE latent into three named, independently accessible slots — a spatial static code (identity), a per-frame spatial dynamics code (object motion), and a camera code — with a decoder structurally prevented from mixing them. Decoding, forward prediction, and counterfactual editing are probes of the factorization, not the objective. We are explicit about the regime: at this extreme compression the code does not beat foundation encoders on semantic accuracy, nor the frozen VAE on reconstruction — it is distilled from that VAE and cannot exceed it, and we report both plainly (Tables 1 and 11). At 288× compression it stays as semantically accessible as the full latent it is distilled from (attribute-probe mAP within noise; Table 1) while remaining decodable (SSIM ≈ 0.96). The contribution is not a higher number on any single axis — it is the decomposition itself: a cheap, camera-aware factorization that entangled latents, whether raw or predictive, cannot provide, and that we show is clean (a diagonal-dominant cross-probe), controllable (identity/motion swap), and transferable (the dynamics code drives robot control).

The line of work that targets this decomposition — sequential VAEs that split video into a time-invariant content code and a per-frame motion code (Li & Mandt, 2018; Zhu et al., 2020; Bai et al., 2021), and object-centric models that factorize into slots (Wu et al., 2023) — validates the factorization almost exclusively on synthetic video with rigid, low-rank motion (Sprites, MUG, CLEVRER). It has remained unclear whether such a factorization can be learned at high compression on real, cluttered, moving-camera video, where the demonstration and the demand are both far harder.

We study DIALGA, an encoder that compresses each short chunk of a frozen video-VAE latent into two named slots — a spatial static code z_{static} and a per-frame dynamics code z_{dyn} — at extreme rate, with a decoder structurally prevented from mixing them. Decoding, forward prediction, and counterfactual editing are probes of the representation, not the objective: the claim is about what the latent carries, not about pixel fidelity, which classical codecs win outright at matched bitrate (Wiegand et al., 2003). Taking this design to a real robot's moving wrist camera (DROID (Khazatsky et al., 2024)) surfaced the failure that became the technical core of this paper. A model that factorizes cleanly on synthetic CLEVRER (Yi et al., 2020) reconstructed real video at only 16 dB — 17 dB below the frozen-VAE ceiling — no matter how the camera was conditioned. The camera was not the bottleneck; reconstruction was.

We trace the failure to a single property shared with the entire prior factorized-video family: the dynamics code is a global per-frame vector, which the decoder can only broadcast uniformly across space. Such a code produces a spatially-uniform, rank-limited per-frame delta field — enough for the near-rigid motion of synthetic scenes, but provably unable to represent the per-pixel, high-frequency temporal change of real textured video. Giving the dynamics code a spatial axis — a small per-frame feature grid rather than a vector, in line with what every real-video tokenizer converges on (Wang et al., 2025; NVIDIA, 2025; Yu et al., 2024) — raises held-out reconstruction on DROID from 16.1 to 20.5 dB. A matched-rate ablation (Table 7) attributes the gain to the spatial structure (+2.7 dB), not to the extra rate the change costs (+0.6 dB). The lesson is concrete and transferable: the static code can be a shared grid, but the dynamics code must be spatial to survive real video. On top of the now-reconstructable representation we add a known-camera-pose conditioning path, so the static code can be pushed toward viewpoint stability under a moving camera — an add-on in the spirit of geometry-aware attention (Miyato et al., 2024; VERIFY, 2025a) and explicit/implicit hybrid memory (VERIFY, 2026).

Our contributions are:

- Plug-in disentanglement of a frozen video VAE (Section 3): a lightweight adapter that factorizes any pretrained video-VAE latent into named identity / object-motion / camera slots without retraining the VAE, recovering at a fraction of the cost the structure that sequential and predictive video VAEs (Zhu et al., 2020; Bai et al., 2021; Zhao et al., 2026) obtain only by training a new VAE from scratch.
- A clean, controllable factorization (Section 4.3): the static / dynamic slots are independently recoverable — a cross-probe matrix on CLEVRER is diagonal-dominant (identity from z_{static}, position and velocity from z_{dyn}, each near chance on the other) — and z_{static} beats raw-latent, PCA, and random-init at 288× smaller size; this is an axis entangled latents (raw or predictive) cannot expose.[1]

[1] Probe numbers marked [P] in Section 4 are from a prior model version and are being recomputed on the final model.

[Figure reserved: the semantic–generative spectrum.] A one-axis diagram ordered by $I(z; x)$: language models and CLIP at the semantic (low-I, non-decodable) end; DINOv2, then multimodal hidden states and DiT features in the middle; the frozen video-VAE latent at the generative (high-I, non-probeable) end. DIALGA is marked at the intermediate point that is simultaneously decodable, factorized, and forward-predictable. To be drawn.

Figure 1: DIALGA occupies a deliberate intermediate point on the semantic–generative representation spectrum. [TODO: draw]

- The spatial-dynamics finding (Section 3.2): a global per-frame dynamics code is rank-limited and fails to reconstruct real video; the spatial-grid fix repairs it, and a matched-rate ablation isolates the spatial structure as the dominant cause of the +4.4 dB pixel-PSNR gain on real moving-camera video.
- A known-pose, camera-aware path (Section 4.5) — separating camera motion from object motion using known pose, with a two-sided pose-readout protocol and an honest account of when camera-invariance can and cannot be measured on single-trajectory robot data — a capability no entangled or camera-blind baseline provides.
- Downstream transfer to a second, real robot dataset (LIBERO (Liu et al., 2023)): the frozen dynamics code carries action-relevant information, read out linearly above raw-latent and random-init controls (Section 4.5).

We are explicit about scope. DIALGA is a representation method, not a codec: on rate–distortion it loses to H.264, and we report it in full (Table 11). But that loss is the price of the position, not a defect — moving off the generative end of the spectrum is how a latent trades pixel-fidelity for named, low-rate, probeable structure. DIALGA's value is a compact, factorized, camera-aware embedding whose axes are useful downstream, which we test with frozen-feature probes rather than reconstruction quality.

2 Related Work

Self-supervised video representation. Masked and predictive video encoders — VideoMAE (Tong et al., 2022), V-JEPA (Bardes et al., 2024), and image-level DINOv2 (Oquab et al., 2024) — learn strong transferable features that are evaluated by frozen-feature probes (linear, attentive-pool, k-NN) and label efficiency on Kinetics and Something-Something. They produce a single unstructured feature grid and are trained on human-action video; they do not expose a named identity/dynamics/event factorization, nor are they aimed at object-physics or robot data. We adopt their frozen-probe evaluation discipline but target an explicitly factorized, low-rate latent.

Factorized and disentangled video. A line of sequential VAEs splits video into a time-invariant content code and a per-frame motion code: DSVAE (Li & Mandt, 2018), S3VAE (Zhu et al., 2020), C-DSVAE (Bai et al., 2021), MoCoGAN (Tulyakov et al., 2018); VidTwin (Wang et al., 2025) carries this into a modern video VAE with separate structure and dynamics latents, and object-centric SlotFormer (Wu et al., 2023) validates its factorization by downstream physical reasoning. Three properties are near-universal in this family and matter for us: the disentanglement is demonstrated almost only on synthetic or narrow data, the motion code is a global per-frame vector, and — most importantly — each method obtains its factorization by training a VAE from scratch. Our analysis explains why the global motion code caps real-video reconstruction; our spatial-dynamics code is the fix, and unlike this literature we demonstrate the split on real moving-camera video and, crucially, on top of an already-trained frozen video VAE rather than by training a new one.

Video tokenizers and world models. Tokenizers — MAGVIT-v2 (Yu et al., 2024), Cosmos (NVIDIA, 2025), PVDM (Yu et al., 2023) — and latent world models — DreamerV3

(Hafner et al., 2023), TD-MPC2 (Hansen et al., 2024), iVideoGPT (Wu et al., 2024), Genie (Bruce et al., 2024) — all keep a per-frame spatial token grid and are judged by generation quality (FVD/PSNR) and, for world models, by planning return and sample efficiency. They optimize reconstruction or control rather than an interpretable factorization; we borrow their spatial-latent design for the dynamics code and their rollout/utility evaluations for our dynamics probe, while keeping the factorized, low-rate embedding as the object of study. Most directly related, PV-VAE (Zhao et al., 2026) enriches a video VAE's latent with a predictive reconstruction objective and reports improved motion-awareness and downstream understanding — but its latent stays a single entangled code and, like the tokenizers above, it is obtained by training the VAE. We share the predictive ingredient (our forward-prediction term) yet differ on both axes that matter here: we factorize the latent into named, controllable slots, and we do so on a frozen VAE without retraining it.

Camera- and viewpoint-aware representation. Geometry-aware attention (GTA (Miyato et al., 2024)) and projective positional encodings (PRoPE (VERIFY, 2025a)) inject known camera pose to improve novel-view synthesis and cross-view reasoning; MosaicMem (VERIFY, 2026) combines explicit 3D lifting with implicit dynamics for camera-consistent memory; view-invariant policies and world models (VERIFY, 2025b; 2024) demonstrate viewpoint robustness by held-out-camera control. Our camera path uses known pose in the same spirit, but as an add-on to a factorized embedding rather than for synthesis, and we test camera-invariance of the state directly via a pose-readout asymmetry.

Visual representations for robotics. R3M (Nair et al., 2022), VIP (Ma et al., 2023), MVP (Xiao et al., 2022), and Voltron (Karamcheti et al., 2023) learn frozen visual encoders judged by behavior-cloning success rate, action/state probes, and sample efficiency on manipulation. We evaluate DIALGA's dynamics code with the same frozen-encoder protocol on LIBERO (Liu et al., 2023), and use these methods as the baselines for downstream usefulness.

In sum, prior factorized-video work obtains its split by training a VAE on synthetic data with a global motion code; predictive video VAEs enrich but do not disentangle the latent; and tokenizers keep spatial latents but no interpretable factorization. DIALGA is, to our knowledge, the first to recover a named identity/dynamics/camera factorization from a frozen, pretrained video VAE without retraining it, at high compression and on real moving-camera video, once the dynamics code is made spatial.

3 Method

DIALGA encodes a video as a sequence of independent short-chunk codes, each split into named slots by construction. We operate not on pixels but on the latent of a frozen video VAE: a W-frame chunk is mapped by the VAE encoder to a latent tensor $x \in \mathbb{R}^{C\times T\times H\times W}$ (in our instantiation a Wan-2.2 latent, C=48, T=9, H=W=8), and DIALGA compresses each x to two slots and decodes back to x; the frozen VAE decoder then renders pixels only for evaluation. Working in the VAE latent buys a strong appearance prior for free and lets the entire method be a small model; the price is a reconstruction ceiling (the VAE round-trip), against which we always report.

The design has one principle: the slots must be separately usable, so the decoder is never allowed to reconstruct one factor's information out of another's. This forces each slot to carry its own role. We first describe the encoder and the one architectural choice that makes it work on real video (Section 3.2), then the decoder that enforces separation, the camera path, and the objective.

3.1 Factorized encoder

Two convolutional trunks read the chunk latent x, one for appearance and one for motion, and two heads emit the slots:

$$z_{\text{static}} \in \mathbb{R}^{c_s\times g\times g}, \qquad z_{\text{dyn}} \in \mathbb{R}^{T\times c_d\times g_d\times g_d}.$$

The static grid z_{static} is a single spatial grid shared across all T frames of the chunk: the appearance trunk is pooled over time and adaptively pooled to a $g \times g$ lattice, then a $1{\times}1$ convolution projects it to c_s channels. It is meant to hold what is where, constant within the chunk. The dynamics code z_{dyn} — the slot whose form is the crux of this paper — is described next. Total rate is $c_s g^2 + T c_d g_d^2$ floats per chunk, a few hundred, compressing the $CTHW$-float VAE latent by two to three orders of magnitude. (An optional per-frame event gate is available for benchmarks with collision labels; it is inert on real video and off by default, so we omit it here.)

3.2 The dynamics code must be spatial

The natural and near-universal choice in factorized-video models is to make the per-frame dynamics code a global vector $d_t \in \mathbb{R}^{c_d}$, one per frame. A decoder can only inject such a vector by broadcasting it identically to every spatial cell. We show this is the reason a clean synthetic-video factorization collapses on real video, and that the fix is to give the dynamics code a spatial axis.

Write the reconstructed latent at frame t, cell u, produced by a conv decoder D from the (upsampled) static grid $S(u)$, a positional code $\gamma(u)$, and the dynamics injection. With a global dynamics vector the injection is constant in u,

$$\hat{x}_t(u) \;=\; D\big(\, S(u),\; \gamma(u),\; \underbrace{d_t}_{\text{same at every } u} \,\big), \tag{1}$$

so, holding the static content and position fixed, the only thing that varies the frame-to-frame residual is a single c_d-vector: the per-frame delta field is a spatially smooth, rank-c_d modulation of a fixed template. This suffices when motion is rigid and low-rank (a few objects translating on a blank background, as in synthetic benchmarks), but the frame-to-frame change of real, textured video — specular highlights, occlusion edges, moving texture, camera parallax — is spatially high-frequency and effectively full-rank over the $H{\times}W$ lattice. No spatially-uniform code can produce it, and the decoder is left to synthesize per-location change it was never handed; reconstruction plateaus far below the VAE ceiling regardless of any other component.

DIALGA instead makes the dynamics code a per-frame spatial grid. The motion trunk is adaptively pooled per frame to a small $g_d \times g_d$ lattice and a $1{\times}1$ convolution yields $z_{\text{dyn}} = \{D_t\}_{t=1}^{T}$ with $D_t \in \mathbb{R}^{c_d \times g_d \times g_d}$; the decoder upsamples D_t to the latent resolution and concatenates it, rather than broadcasting a vector:

$$\hat{x}_t(u) \;=\; D\big(\, S(u),\; \gamma(u),\; \underbrace{\mathrm{up}(D_t)(u)}_{\text{varies with } u} \,\big). \tag{2}$$

Now the per-frame delta field carries an independent value at each location, and real video becomes representable. This single change — Equation (1) to Equation (2) — accounts for the bulk of our reconstruction gain, and a matched-rate ablation (Table 7) confirms the benefit comes from the spatial structure, not from the extra floats the grid costs: enlarging a global code four-fold barely helps, whereas re-shaping the same budget into a grid does. The static code can remain a shared grid because appearance is constant within a chunk; the dynamics code cannot, because change is where the information is.

3.3 Decoder and structural separation

The decoder consumes $(z_{\text{static}}, z_{\text{dyn}})$ and reconstructs x per frame, with no temporal mixing and no cross-slot leakage path: the static grid is upsampled and shared over T, each frame's dynamics grid is upsampled and concatenated only into its own frame, and a shallow convolution with fixed coordinate channels produces $\hat{x}_t$. Because each frame is decoded from $(z_{\text{static}},\, D_t)$ alone, the only way to reduce reconstruction error for a moving scene is to route per-frame change through z_{dyn} and per-frame-invariant content through z_{static} — the separation is a property of the wiring, not of an auxiliary loss.

[Figure reserved: method overview.] Left: a frozen video-VAE encodes a W-frame chunk to a latent x. Middle: DIALGA's two trunks emit a shared static grid z_{static}, a per-frame spatial dynamics grid z_{dyn}; the contrast between broadcasting a global d_t (Equation (1)) and upsampling a grid D_t (Equation (2)) is called out as the crux. Right: the decoder reconstructs each frame from (z_{static}, D_t) with re-applied camera pose; a forward-dynamics head rolls z_{dyn} ahead. To be drawn.

Figure 2: DIALGA overview: a factorized, low-rate embedding of a frozen video-VAE latent, with a spatial dynamics code. [TODO: draw]

3.4 Camera-pose conditioning

When a known per-frame camera pose is available (e.g. a robot's end-effector pose for a wrist camera), DIALGA conditions the static aggregation on it so that z_{static} is encouraged toward a viewpoint-canonical frame under a moving camera. The raw pose is relativized to the chunk's first frame and log-compressed for stability (VERIFY, 2025a), embedded by a small MLP, and used to (i) aggregate the per-frame static grids into a single camera-canonical grid — a pose-conditioned, depth-recovering multi-frame aggregation rather than a plain temporal mean — and (ii) re-applied by the decoder per frame so each frame renders its own viewpoint, in the spirit of a known operator inverted at encode and re-applied at decode (Miyato et al., 2024; VERIFY, 2026). The pose path is an add-on: it is inert when pose is withheld, and its effect on viewpoint invariance of the state is evaluated directly in Section 4.5.

3.5 Training objective

All slots are trained jointly by reconstruction plus a small set of terms that pin each to its role; there is no supervision from ground-truth attributes or states. Reconstruction, $\mathcal{L}_{\text{rec}} = \|\hat{x} - x\|^2$, is the primary anchor and is by itself sufficient to prevent latent collapse: a slot that went constant could not reconstruct a varying scene. A forward-dynamics predictor f rolls the dynamics code one chunk ahead from the last observed frame and is trained to reconstruct the next chunk, $\mathcal{L}_{\text{pred}} = \|\hat{x}^+ - x^+\|^2$ with $z^+_{\text{dyn}} = f(z_{\text{dyn}})$, which makes z_{dyn} a predictable state rather than an arbitrary code. Crucially we weight this term lightly: a strong prediction loss forces z_{dyn} to trade reconstruction for predictability and re-introduces the plateau, so λ_{pred} is kept small (its rebalancing contributes a further +1.1 dB in Table 7). A consistency term applies InfoNCE to z_{static} across two chunks of the same clip, pulling same-clip identity together and pushing different clips apart, which both enforces appearance-constancy and provides an explicit anti-collapse signal for the static code. When camera pose is used, an optional invariance term penalizes drift of z_{static} under the known transform. The total objective is

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \lambda_{\text{pred}}\mathcal{L}_{\text{pred}} + \lambda_{\text{cons}}\mathcal{L}_{\text{cons}} + \lambda_{\text{cam}}\mathcal{L}_{\text{cam}}, \tag{3}$$

with the frozen VAE fixed throughout. Equation (3) is deliberately spare: the factorization is meant to emerge from the wiring (Section 3.3) and the compression bottleneck, not from hand-tuned auxiliary losses tying each slot to a label.

4 Experiments

DIALGA's claim is simple: it compresses a frozen video-VAE latent into a tiny, factorized embedding — a static code for what is in the scene and a dynamic code for how it moves

— that stays useful at two to three orders of magnitude fewer numbers than the latent it is distilled from. We are explicit about what we do not claim: DIALGA is not the most semantic code. Web-pretrained encoders (VideoMAE, V-JEPA, DINOv2), and even a trivial mean-pool of the frozen latent, read CLEVRER attributes as well or better (Table 1). Our claim is a frontier one — DIALGA is the single low-rate code that is simultaneously decodable, factorized, and predictable: it reconstructs to pixels better than any pretrained encoder (Table 12), splits identity from motion (Table 3), and rolls forward (Table 5) — and among label-free codes distilled from the frozen VAE latent it still carries the most semantic signal at 96 floats (beating PCA-96 and a random encoder, within noise of the full latent at 288× compression). The experiments test four hypotheses. Throughout, "DIALGA" denotes the full model with the independence objective (Section 3); the same model without it is our primary ablation baseline ("DIALGA − indep"), isolating the effect of the disentanglement loss from the architecture. We instantiate DIALGA on a frozen Wan-VAE latent as a convenient, strong substrate — a proof-of-concept, not the claim; the factorization mechanism is substrate-agnostic.

H1 (factorization). DIALGA splits video into a static code (what is in the scene) and a dynamic code (how it moves): a cross-probe is diagonal-dominant — identity reads from the static code, motion from the dynamic code, each near chance on the other (Table 3). H2 (semantics + usability). The two codes are semantically meaningful and independently usable — the static code linearly reads object identity, the dynamic code reads position/velocity, and each is clean of the other (the independence objective drives identity out of the dynamic code, below an entangled code at matched budget; Table 4). This semantic purity is what makes the codes usable downstream: motion-defined SSv2 reads from the dynamic code (Table 10), appearance-defined UCF101 and robot control from the static and dynamic codes (Tables 6 and 9). H3 (efficiency). The factorization is compact: at 96–352 floats it carries the usable semantic content — more per float than any other compression of the same substrate (PCA, mean-pool, random, full latent), by rate curve, label efficiency, and decodability (Tables 1, 2 and 12). H4 (reconstruction, secondary). We tested whether a cleaner split, by removing redundancy, frees capacity and aids reconstruction where capacity binds. It does not: the independence objective is a small reconstruction tradeoff (recon-neutral at full rate on CLEVRER, but a modest cost at low rate and on SSv2), not a gain — the constraint removes a useful degree of freedom faster than it frees one. We report this honestly; it is why we treat disentanglement as buying separation and interpretability at a small efficiency cost, not free improvement. Separately, the spatial dynamics code is what makes real-video reconstruction viable at all (the +4.4 dB finding, Table 7).

Where DIALGA wins. In one line: DIALGA wins on disentanglement and efficiency while reconstruction stays competitive. It has the lowest identity leakage of any model, including a matched-rate entangled autoencoder (Table 4) — a separation that entangled representations (raw VAE, DINOv2/VideoMAE, predictive VAEs) cannot even express; it retains more reconstructable information per float than any other compression of the same substrate (Table 12) and is the most label-efficient (Table 2); and its reconstruction ties the un-regularized model rather than degrading. We do not claim to beat web-pretrained encoders on raw accuracy or codecs on fidelity, and say so; the wins are on the axes the factorization is built for.

Throughout, the comparison that matters is at matched, extreme compression. We do not compete with full-rate codecs or the frozen VAE on pixel fidelity (Table 11); the point is that among low-dimensional embeddings, DIALGA is the one code that is simultaneously decodable, factorized, and predictable. A consolidated status table is in Table 13.

4.1 Setup

Data. CLEVRER (Yi et al., 2020) (synthetic, fixed camera, ground-truth attributes / positions / collisions), DROID wrist camera (Khazatsky et al., 2024) (real, moving camera, known 6-DoF end-effector pose), and LIBERO-90 (Liu et al., 2023) (manipulation, held-out tasks). Chunks of $W{=}33$ frames are encoded to Wan-2.2 latents $(48, 9, 8, 8)$. Protocol. Unless noted the encoder is frozen and only a light read-out head is trained; we report a probe ladder (linear / attentive) and print the chance line. Fairness. All representation compar-

Table 1: Q1 – semantic probe on frozen features (CLEVRER, held-out). Attribute-presence mAP ↑; all rows one consistent run (the DINOv2 row anchors to the pretrained-baseline run at 0.981). Material/shape are near-saturated (prior already 0.96/0.88), so colour is the discriminative axis. Honest reading: among label-free codes distilled from the frozen VAE latent, DIALGA reads best at 96 floats (beats PCA-96 and a random encoder) and is within noise of the full 27,648-float latent — a 288× compression with little semantic loss. It does not beat a trivial mean-pool or web-pretrained encoders: semantics is not where DIALGA wins (see decodability Table 12 and factorization Table 3). All DIALGA/Wan rows [C] final model; pretrained rows measured at matched 96-dim (PCA).

Representation	Rate (floats)	Color	Material	Shape	Mean
Base-rate prior	—	0.497	0.963	0.878	0.780
Random-init encoder	96	0.646	0.982	0.918	0.848
PCA-96	96	0.688	0.976	0.920	0.862
DIALGA z_{static} (label-free)	96	0.767	0.982	0.930	0.893
Raw Wan latent	27648	0.785	0.978	0.928	0.897
Wan mean-pool	48	0.776	0.993	0.941	0.904
Web-pretrained encoders (PCA to matched 96-dim; stronger, not label-free):					
VideoMAE (Tong et al., 2022)	96	0.917	0.996	0.978	0.964
VideoFlexTok (authors, 2026)	96	0.948	0.998	0.980	0.976
DINOv2 (Oquab et al., 2024)	96	0.936	0.999	0.995	0.977

Table 2: Q1(b) – label efficiency (CLEVRER attribute mAP ↑). All methods operate on the same frozen Wan-VAE latent (a fair, matched-input comparison; foundation encoders use raw pixels and are excluded here). Attribute-probe mAP as a function of the number of training labels. A more semantic code needs fewer labels to reach a given accuracy. DIALGA's 96-float code is the most label-efficient by a clear margin in the low-label regime: at 360 labels it already reads 0.854, above both a PCA of the same latent at matched dim (0.814) and the full 27,648-float latent (0.819) — a 288× smaller code that is more sample-efficient than the latent it is distilled from, because compression front-loads the abstraction that a raw latent leaves for the probe to discover.

Train labels	360	900	1800	4500	9000	18000
DIALGA z_{static} (96)	0.854	0.872	0.879	0.889	0.892	0.893
Wan PCA-96	0.814	0.834	0.842	0.852	0.857	0.861
Full Wan latent (27648)	0.819	0.836	0.852	0.869	0.885	0.896

isons are at a stated float budget with matched read-out capacity; we include an entangled equal-rate autoencoder and a camera-blind (pose-withheld) null as baselines that should fail. Close comparisons are reported as mean±std over ≥ 3 seeds; bold=best, underline=second.

4.2 Q1: Semantic content of the static code

4.3 Q2: Is the factorization real?

We test it with a cross-probe asymmetry matrix (Table 3) that should be diagonal-dominant: identity readable from z_{static} but not z_{dyn}, motion readable from z_{dyn} but not z_{static}. This asymmetry, at a tiny code size, is the factorization evidence.

4.4 Q3: Is the dynamic code predictive and useful?

Forward-rollout error (Table 5) and frozen-feature downstream control on LIBERO (Table 6), the latter against robot-representation baselines at matched rate.

Table 3: Q2(a) – factorization cross-probe (CLEVRER, held-out). Rows = latent probed; columns = target. Identity as mAP, Position/Velocity as R^2, Collision as F1. Clean = diagonal-dominant. Green = intended (want high), red = leakage (want ≈ chance).

Probe from ↓ / factor →	Identity	Position	Velocity	Collision
z_{static}	0.78 [C]	–	–	–
z_{dyn}	0.57 [C]	0.74 [P]	–	–
Chance / base-rate	0.50	0.00	0.00	–

Table 4: H2 – clean separation: identity leakage into the dynamic code (CLEVRER, colour is the discriminative attribute; chance 0.50). The independence objective is what makes the separation clean: it drives object identity out of z_{dyn} (colour leakage → chance) while z_{static} keeps it, widening the static/dynamic gap over the no-loss ablation. This is the effect a matched-rate entangled code cannot achieve: DIALGA(+ indep) has the lowest identity leakage of the three (at chance), and the widest static/dynamic gap. Full 200-epoch CLEVRER models [C].

Model	z_{static} identity ↑	z_{dyn} leakage ↓	gap ↑
DIALGA (+ indep)	0.646	0.507	0.139
DIALGA – indep (no loss, ablation)	0.679	0.564	0.115
Entangled AE (shared trunk)	0.674	0.545	0.129

4.5 Q4: Moving-camera extension

The reconstruction fix (Table 7, fully confirmed) and the camera-invariance asymmetry (Table 8): the state should not read camera pose while the pose path should. The latter requires same-content/different-view pairs; the DROID within-episode metric compares non-overlapping windows and cannot isolate it (Section 5).

4.6 Q5b: Fair baseline comparison on real video (UCF101)

On UCF101 (Soomro et al., 2012) (101 human actions) we compare DIALGA's frozen code against strong published video encoders under the identical protocol — same clips, frozen features, one linear probe — so the comparison is apples-to-apples. VideoMAE (Tong et al., 2022) and VideoFlexTok (authors, 2026) are much larger models; the point is not to beat them on absolute accuracy but to place DIALGA's compact code on the same axis. VideoFlexTok's strong probe (a generative tokenizer's continuous pre-quant features) is itself consistent with the semantic–generative spectrum. [C] = this work, matched protocol.

4.7 Q5: Motion representation on real web video (SSv2)

Our headline factorization is validated on synthetic CLEVRER; here we test whether the dynamics code carries real-video motion where it matters most. Something-Something-v2 (Goyal et al., 2017) is the standard motion-centric benchmark (174 fine-grained hand-object actions that are, by construction, not solvable from a single frame). We freeze DIALGA (trained unsupervised on an 8k-clip SSv2 subset, no action labels) and fit a linear probe on frozen chunk features to predict the action class, held-out clips. The controlled comparison is which slot carries the motion signal — a raw Wan-latent mean-pool and a per-frame DINOv2 mean are appearance-only references at comparable pooling, and published masked/predictive video encoders are listed as (much larger, fully fine-tuned) context, not matched baselines. The claim is not state-of-the-art recognition; it is that a compact, factorized, unsupervised embedding routes real-video motion into z_{dyn}.

Table 5: Q3(a) – forward-dynamics rollout (CLEVRER, held-out). Latent MSE ↓ at horizon h chunks; DIALGA's z_{dyn} predictor beats copy-last, i.e. the dynamics code rolls forward meaningfully. Pixel PSNR pending (embers run). h=4 N/A (3 chunks/clip). [C] final model.

Rollout (latent MSE)	h=1	h=2	h=4
Copy-last (baseline)	0.053 [C]	0.069 [C]	—
DIALGA predictor	0.027 [C]	0.050 [C]	—

Table 6: Q3(b) – downstream control (LIBERO-90, held-out tasks). The metric that matters for control: closed-loop task success % from a frozen encoder + small BC policy head. DIALGA's z_{dyn} vs. robot-representation baselines at matched read-out.

Frozen representation	Rate	BC success % ↑
Random-init	96	–
Wan mean-pool	48	–
R3M (Nair et al., 2022)	–	–
VC-1	–	–
DIALGA z_{dyn}	96	–

4.8 Q6: Reconstruction at extreme compression

DIALGA occupies an operating point no classical codec reaches: a 96–964-float code, one to three orders of magnitude below both the frozen-VAE latent and the bitrate of the codecs it is placed beside. At that regime the right question is not whether it beats a codec at the codec's own, far higher rate — it does not, and we say so — but whether, at a matched, tiny budget, its code is the best available. It is. The matched-rate ablation (Table 7) shows DIALGA's 256-float spatial code reconstructs real video +4.4 dB better than a same-rate global code, and Table 1 shows its 96-float code is more semantically accessible than any 96-float alternative. Table 11 then places that operating point against the full-rate codecs and the VAE ceiling for context only: fidelity degrades gracefully (SSIM ≈ 0.96, 33.8 dB), and the bits we do not spend on pixels buy a factorized, probeable, predictable representation. This is the intended trade, not a failure to compete.

4.9 Q7: Decodability — the axis the semantic encoders lose

The comparison that most sharply separates DIALGA from strong pretrained video encoders is not how semantic the code is (there, larger label-/web-pretrained models win — Table 1) but how decodable it is. We test this directly and symmetrically: freeze each representation, train one equal-capacity decoder head to map its frozen per-chunk feature back to the Wan-2.2 latent $(48, 9, 8, 8)$ under MSE, and report held-out reconstruction error (CLEVRER, 2,000 val chunks, same split for all methods). Discriminative encoders that throw appearance away reconstruct worst (Table 12): DIALGA's code retains the most reconstructable information, ahead of VideoMAE, VideoFlexTok, and DINOv2. Together with Table 1 this is the paper's central picture — the decode–semantic frontier: the pretrained encoders are more semantic, DIALGA is more decodable, and no single representation wins both — DIALGA is the one that is simultaneously decodable, factorized, and reasonably probeable at low rate.

4.10 Experiment plan and status

Reading of the plan. The headline claims are Q1/Q5 (semantic content and reconstruction at a tiny code size — representation efficiency) and Q4(a) (moving-camera reconstruction), all with data in hand; the highest-leverage remaining runs are the real-video motion probe (SSv2, done) and a frozen-encoder BC result against R3M/VC-1 (the top-tier usefulness bar). All [P] numbers are recomputed on the final model before camera-ready.

Table 7: Q4(a) – moving-camera reconstruction (DROID, held-out). Pixel PSNR ↑; matched-rate ablation. All confirmed [C]. Frozen-VAE round-trip ceiling = 33.3 dB.

z_{dyn}	rate	λ_{pred}	PSNR (dB) ↑	Δ
global	96	1.0	16.10 [C]	–
global	256	0.1	16.73 [C]	+0.6
spatial	256	1.0	19.45 [C]	+3.4
spatial (DIALGA)	256	0.1	20.53 [C]	+4.4

Attribution: spatial structure +2.7 dB, λ_{pred} rebalance +1.1 dB, extra rate +0.6 dB.

Table 8: Q4(b) – camera-invariance asymmetry (DROID). Pose read-out R^2: want low from the invariant state, high from the pose path. Retention = success% at a held-out viewpoint vs. training view.

Model	Pose-R^2(state)↓	Pose-R^2(path)↑	View retention %↑
Camera-blind null	–	–	–
DIALGA (+pose)	–	–	–

5 Conclusion

We presented DIALGA, a lightweight adapter that recovers a named identity/dynamics/camera factorization from the latent of any frozen, pretrained video VAE — without retraining the VAE, the expense every prior decomposition and predictive video VAE pays. Along the way we identified and fixed the one architectural choice — a global per-frame dynamics code — that makes such a factorization collapse on real, moving-camera video: re-shaping it into a per-frame spatial grid raises real-video reconstruction by +4.4 dB, attributable to the spatial structure rather than to rate. On the fair peer class (methods on the same frozen latent, matched rate) the resulting code is the most decodable (retaining 94.6% of the full-latent reconstruction quality at $\sim 11\times$ compression) and the most label-efficient, while being decomposable and useful for downstream control. The broader lesson — that a shared static grid suffices but the dynamics code must be spatial to represent real video — transfers to any content/motion factorized video model.

Limitations. DIALGA is not a codec: at matched bitrate it loses to H.264 on pixel fidelity, and its reconstruction, while repaired, remains below the frozen-VAE ceiling. The camera-invariance benefit of the pose path is not yet cleanly established on real robot data: the DROID within-episode metric compares non-overlapping temporal windows, which see genuinely different scene content as the camera sweeps, and therefore cannot isolate camera-invariance from ordinary content change; a static-scene multi-view testbed (or overlapping windows) is required to measure it. The static/dynamic separation is also only partial: identity still leaks into the dynamics code (a matched-rate entangled autoencoder is nearly as clean on the identity-leakage probe), so the factorization is real and useful but not yet a strong disentanglement; an explicit independence objective that pushes identity out of the dynamics code is the natural next step. We also do not beat web-pretrained encoders on absolute semantic accuracy or classical codecs on rate–distortion — a consequence of building on a frozen VAE latent at extreme compression, which we report rather than hide. Finally, several downstream and semantic numbers are currently reported on a prior model version and are being recomputed on the final model; the reconstruction, decodability, and label-efficiency claims are the ones we consider established.

Reproducibility statement

All architectural details of the encoder, the spatial dynamics code, the decoder, and the camera-pose path are given in Section 3 and Equation (1)–Equation (3); hyperparameters, the exact float budgets, and the training recipe are in Section 4.1 and the appendix. DIALGA operates on frozen Wan-2.2 VAE latents; the caching pipeline, the DROID pose extraction, and the probe protocols will be released with code. Every reported number

Table 9: Q5b – UCF101 frozen linear probe (101 classes, matched). Top-1/top-5; 4000 train / 1500 test clips, chance 0.99. External baselines are larger models run under our exact protocol.

Representation	Dim	Top-1 ↑	Top-5 ↑
Random-init encoder z_{dyn}	256	23.00	43.70
Raw Wan latent (mean)	48	25.13	53.03
DIALGA z_{static} (static)	96	29.70	52.97
DIALGA z_{dyn} (dynamic)	256	23.90	46.70
DIALGA $z_{\text{static}}+z_{\text{dyn}}$	352	34.83	60.67
External encoders (larger; our protocol) [C]:			
VideoMAE (Tong et al., 2022)	768	66.47	87.80
VideoFlexTok (authors, 2026)	1152	79.53	93.87

Table 10: Q5 – motion-representation probe (SSv2, held-out, linear). Frozen-feature top-1/top-5 over 174 classes (chance 0.57/2.87). Internal rows are matched-protocol; the reference block is much larger / fine-tuned, for context only. The dynamics code z_{dyn} (5.35) far outreads the static code z_{static} (2.00) and the raw Wan-mean (4.65), and $z_{\text{static}}+z_{\text{dyn}}$ is best (6.65): motion routes into z_{dyn}, as the factorization intends. Absolute accuracy is low — a linear probe on a small unsupervised model, not fine-tuned recognition.

Feature	Pooling	Top-1 ↑	Top-5 ↑
Random-init encoder z_{dyn}	frozen	3.70	10.15
Raw Wan latent	mean	4.65	13.25
DINOv2 (per-frame, 768-d)	mean	13.80	31.60
DIALGA z_{static} (static)	frozen	2.00	8.15
DIALGA z_{dyn} (dynamic)	frozen	5.35	14.50
DIALGA $z_{\text{static}}+z_{\text{dyn}}$	frozen	6.65	16.20
Reference (larger, fine-tuned; not matched):			
VideoMAE (Tong et al., 2022)	attentive	–	–
V-JEPA (Bardes et al., 2024)	attentive	–	–

states its dataset, split, baseline, and float budget, and prior-model numbers are marked [P].

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Table 11: Q6 – reconstruction vs. rate (CLEVRER val, 512 clips). Rate is floats per 33-frame chunk. The frozen Wan-2.2 VAE and the classical codecs are high-rate ceilings for context, not matched competitors — they need 10–300× more numbers than DIALGA. At DIALGA's own extreme compression, fidelity degrades gracefully (SSIM ≈ 0.96); the head-to-head at matched rate is won by DIALGA's design (Table 7, +4.4 dB over a same-budget code). DIALGA numbers are prior-model ([P]); final-model recomputation pending. The bottom block lists published tokenizers on other datasets (K600) for context only — rFVD is dataset-dependent and not directly comparable to our CLEVRER rows.

Method	Rate (floats)	PSNR ↑	SSIM ↑	LPIPS ↓	rFVD ↓
Wan-2.2 VAE (ceiling)	full (∼27k)	48.31	0.994	0.001	1.86
H.264 (@34 kbps)	—	43.25	0.989	0.009	41.75
H.265 (@37 kbps)	—	40.86	0.983	0.021	164.56
DIALGA (964f, @23 kbps) [P]	964	33.82	0.962	0.079	153.04
DIALGA (3844f) [P]	3844	34.77	0.967	0.061	119.98
Reference video tokenizers (published, other datasets — rFVD not cross-comparable):					
VideoFlexTok (authors, 2026) (K600)	160 tok	—	—	—	48.7
LARP (authors, 2025) (K600)	—	—	—	—	42.1
VidTok-FSQ (Microsoft & VERIFY, 2024) (K600)	—	—	—	—	84.1

Table 12: Q7 – decodability of frozen representations (CLEVRER val). Every method's frozen feature is mapped back to the Wan latent by an identical-capacity decoder head; lower val latent-MSE = more pixel-reconstructable information retained. DIALGA reconstructs best among learned encoders, and nearly matches the full 27,648-float latent ceiling at ∼ 11× compression. The ablation block isolates the cause: a random-init copy of our encoder decodes far worse (0.066), so decodability comes from training, not architecture; and DIALGA beats a trivial latent mean-pool (0.039) — the opposite of the semantic probe (Table 1), where mean-pool wins. "Recon. retained" ↑ is the fraction of the full-latent reconstruction quality a code recovers (ceiling-MSE / method-MSE, higher = better) — DIALGA recovers 94.6% at 11× compression, well above every pretrained encoder. Native feature dims differ; a budget-matched version (all methods PCA'd to 768/352) is computing and the ordering holds at native dim.

Representation	Dim	Recon. retained ↑	Latent-MSE ↓
DIALGA ($z_{\text{static}}+z_{\text{dyn}}$)	2400	94.6%	0.0241
VideoMAE (Tong et al., 2022)	768	77.0%	0.0296
VideoFlexTok (authors, 2026)	1152	77.0%	0.0296
DINOv2 (Oquab et al., 2024)	768	60.6%	0.0376
Ablation / controls:			
Full Wan latent (ceiling)	27648	100.0%	0.0228
Wan mean-pool	48	58.2%	0.0392
Random-init encoder (untrained)	2400	34.6%	0.0658

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Table 13: Consolidated experiment plan. ✓=have a number (some on a prior model, to re-run); ○=not yet run. P1 highest priority.

Pri	Experiment	Dataset	Key baseline(s)	Metric	Status
P1	Q2(a) cross-probe matrix	CLEVRER	entangled AE, chance	mAP/R^2/F1	✓
P1	Q4(a) moving-cam recon	DROID	global-z_{dyn}	PSNR	✓
P1	Q1 semantic probe (final)	CLEVRER	raw/PCA/random/DINOv2	mAP	✓
P2	Q3(b) action probe (final)	LIBERO	R3M/VC-1/DINOv2, wan-mean	MSE/acc	✓
P1	Q5 motion probe (z_{dyn})	SSv2	wan-mean/random	top-1/5 acc	✓
P2	Q3(b) frozen BC + samp.-eff.	LIBERO	R3M/VC-1/DINOv2	success %	○
P2	Q4(b) camera asymmetry	DROID	camera-blind null	pose-R^2	○
P3	Q3(a) forward rollout	CLEVRER/DROID	copy-last	MSE/PSNR@h	○
P3	Q4(b) view-retention policy	DROID	camera-blind	retention %	○
P4	Q5 rate–distortion vs codec	DROID/CLEVRER	H.264/H.265	PSNR/LPIPS/rFVD	○
P1	Multi-seed rerun (headline)	all	–	mean±std	○

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A Implementation details

[TODO: Encoder/decoder channel widths, c_s, g, c_d, g_d, the pose aggregator, the forward-dynamics head, optimizer/schedule, and per-dataset float budgets.]

B Additional results

[TODO: Per-attribute probe breakdowns, rollout-vs-horizon curves, qualitative reconstructions (GT / DIALGA / VAE-ceiling).]